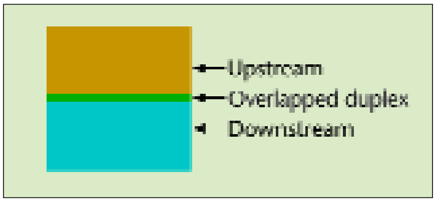
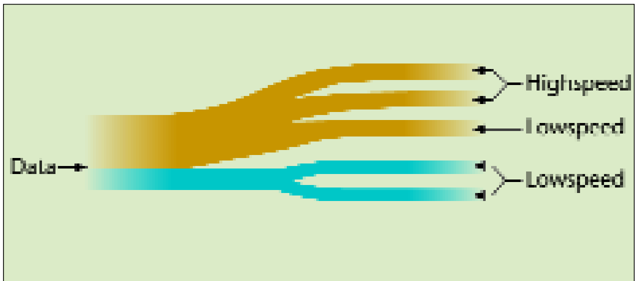


***Echo cancellation:** The available bandwidth is divided into upstream and downstream, and these overlap to create a duplex band*



***Frequency Division Multiplexing:** The available bandwidth is divided into upstream and downstream, without any overlap. All bands are multiplexed on the same line*

company (typically a Digital Subscriber Line Access Multiplexer, or DSLAM) and the other at the subscriber's home, are DSL modems. DSLAM is a mechanism at a phone company's central location that links many customer DSL connections to a single high-speed ATM line.

When the phone company receives a DSL signal from the subscriber, an ADSL modem combined with a POTS splitter detects voice calls and data signals. Voice calls are sent to the Public Switched Telephone Network subscriber line, or PSTN, and data signals are sent to the DSLAM, where it passes through to the Internet, then back through the DSLAM and ADSL modem before returning to the subscriber's PC.

The two separate lines—one from the external PSTN network and the other from the broadband service provider—are merged using a pair of filters, one of them a high pass filter (HPF), the other one a low pass filter (LPF). The data line is first passed through what is called an ATU, and then through the HPF (a reverse of the splitter that is installed at the user's end). Similarly, the POTS line is passed through the LPF. Then they are merged and fed into the 'last mile'. On the subscriber's end, there is a mirror image of the arrangement before the data is fed into the respective machines—voice to telephone, data to computer.

The line connecting them, the 'last mile' as it is referred to, is part of the DSL circuit.

Interestingly, the 'upstream' data rate, or data coming from the client computer to the server, is quite low. The bandwidth requirement is asymmetric. This is what an ADSL or Asymmetric/Asynchronous Digital Subscriber Line circuit does—it suits an asymmetric flow of data needs.

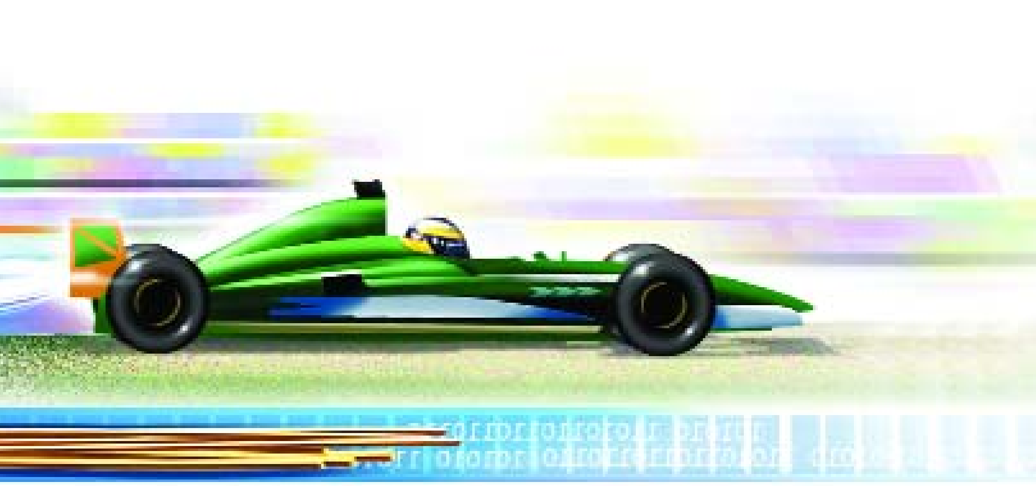
Last mile blues

Most telecom companies, or telcos, use fibre optic technology to communicate with each other. With the passage of time, most developed countries, even some developing ones, have come to possess incredible amounts of backbone bandwidth. The progress of both wire-line and wireless technologies mean that there are effectively no problems in transmitting high bandwidth data from the servers of an Internet Service Provider to the telco.

The bottleneck is created when the broadband data is transferred from the backbone to the much slower Public Switched Telephone Network (PSTN) or, Plain Old Telephone Service (POTS)—the 'last mile'. The last mile comprises the wiring that starts from the telecom service provider to the consumer's home. This section, or 'subscriber loop plant', not only varies widely in terms of quality, but also contains voice-line-specific artefacts installed by telecom companies. The idea is to help voice transmission.

The POTS network, though extremely large and intricate—India has about 43 million fixed-line telephones—is mostly of rather poor quality, and almost entirely consists of 'untwisted' copper wire. Though some countries have recently replaced this in urban or high-demand regions with fibre-optic cables and T1 lines, most of the world's POTS networks still run on age-old wires.

Any T1 system handles POTS voice channels at about 4 KHz for a channel capacity of about 64 Kbps. Common PSTN wiring does, in fact, support much higher bandwidths. However, since the system was set up for voice communication only, all modification to the wiring was done to optimise voice transmission. These include loading coils (originally installed on long circuit loops to cancel noise during voice calls), over long loops (which attenuate signals),



Multiplexers and multiplexing

Multiplexing connects multiple remote sites to a single centralised location. Typically, a connection originating at the host location is fed into a multiplexer at the service provider's end. The multiplexer splits the host circuit into smaller circuits, which are then delivered to the remote sites.

It can be either code-division multiplexing, or time-division multiplexing. In code-division multiplexing, the data stream is divided according to the requirement. In time-division, data is divided according to priority—data that is more important is transmitted faster.